Comparative Study & Practical Analysis of Generative AI Models

Autoencoder • VAE • DCGAN • Stable Diffusion

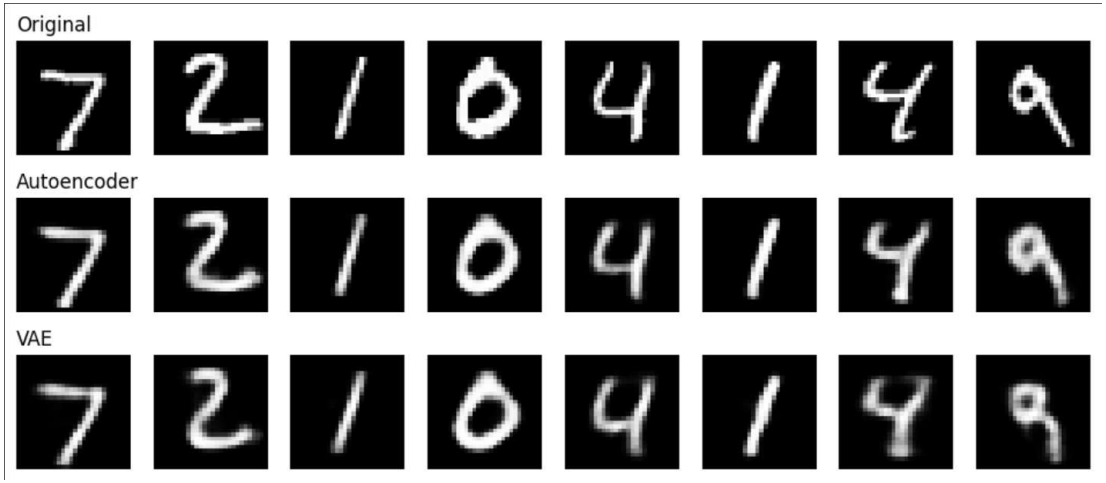
Assignment–02 | CO3 & CO4

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Comparative Study of the Four Models

	Autoencoder	VAE	DCGAN	Stable Diffusion
Principle	Compress, then reconstruct	Probabilistic latent (μ, σ) + KL loss	Generator vs discriminator	Reverse noising, text-guided
Components	Encoder, bottleneck, decoder	Encoder → μ, σ → sample → decoder	Transposed-conv G, strided-conv D	CLIP, U-Net, VAE, scheduler
Advantage	Simple, close reconstruction	Smooth latent, can generate	Sharp, fast sampling	Best quality, text control
Limitation	Cannot generate well	Blurry output	Unstable, mode collapse	Slow, GPU-heavy, artefacts
Applications	Compression, denoising	Generation, anomaly detection	Augmentation, faces	Text-to-image, inpainting
Output	Near-copy of input	Smoother, blurrier	Sharp, low-res, some flaws	Photoreal, high-res

Activity A – Image Reconstruction (CO3)



- AE: sharper and closer to the input; random latent vectors decode to meaningless images.
- VAE: smoother but blurrier, since the KL term trades detail for a regular latent space; random latents decode to plausible digits.
- Reconstruction MSE: AE [0.01149763073772192] vs VAE [0.01557916309684515] (from notebook).

Activity B – Image Generation (CO4)

- DCGAN: sharp digit-like images in one pass, but low resolution, uneven quality and some distorted samples; no text control.
- Stable Diffusion: detailed, high-resolution, prompt-matched images; limits are slow sampling, GPU need, and errors in hands and text.




```
prompts = ['an astronaut riding a horse on the moon, digital art',
           'a photo of a golden retriever puppy playing in a park, high detail',
           'a handwritten digit 7 in black ink on white paper',
           'close-up of a person\'s hands typing on a keyboard']   # last one tests known weak spots (hands)
```

AI Usage, Learning and Contributions

AI tools & prompts

[Tools you used]. Prompts: compare the 4 models; write Colab code for AE/VAE, DCGAN and Stable Diffusion; explain VAE blur.

Verification

Checked claims against original papers (Kingma, Goodfellow, Radford, Ho, Rombach); ran every notebook ourselves; matched text to our screenshots.

Key learning

Only the VAE gives a usable latent space; GANs trade stability for sharpness; diffusion gives best quality at highest compute cost.

Contributions

- [Member 1]: AE and VAE comparison, Activity A
- [Member 2]: DCGAN study and Activity B (DCGAN)
- [Member 3]: Stable Diffusion, prompts, AI-usage log, report